178058: biotite monzogranite, No. 1 Bore

Location and sampling

SANDSTONE (SG 50-16), LAKE MASON (2842)
MGA Zone 50, 783150E 6923390N

Sampled on 30 October 2002

The sample was taken from the edge of a 1.5 m-high, 3 m-diameter boulder on the southern side of a large rocky pavement, 2 km north of No. 1 Bore.

Tectonic unit/relations

This sample is a pale-grey, medium- and even-grained, massive biotite monzogranite from the Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Wyche, 2004). It was collected to determine the age of monzogranite away from major shear zones.

Petrographic description

The principal minerals are plagioclase (50–55 vol.%), K-feldspar (25 vol.%), quartz (20 vol.%), biotite (2–3 vol.%), and opaque oxide (<1 vol.%), with trace amounts of accessory chlorite, epidote, carbonate, muscovite, clay, sericite, fluorite, ?allanite, leucoxene, titanite, and zircon. Anhedral to subhedral plagioclase and anhedral microcline grains are mostly less than 3 mm in length, with perthitic and antiperthitic textures preserved locally. The plagioclase is dominantly oligoclase, with some possible andesine and albite, the latter exsolved from adjacent microcline. Some plagioclase grains contain large, poorly defined patches of myrmekite. Partly poikilitic quartz grains reach 6 mm in length and have weak undulose extinction. The biotite, up to 1 mm in grain size, is unoriented and accompanied by granular opaque oxide as single grains and small lenses. A single patch of purple fluorite is less than 1 mm in diameter. Zircons are small (<50 µm in diameter) and mostly attached to opaque oxide grains. Rare, small clouded grains could be altered zircon, but are more probably altered allanite. Titanite is rare. Relatively coarse grained alteration products in the plagioclase include muscovite, calcite, and epidote, developed in various combinations, with later clay and sericite in irregular patches. Chlorite and epidote have replaced a small proportion of the biotite, with leucoxene or microcrystalline titanite resulting from the replacement. Late carbonate-filled microfractures could be supergene. The fabric of the granitoid suggests minimal deformation and metamorphism, with low-temperature muscovite–epidote–chlorite–carbonate and clay–sericite assemblages.

Zircon morphology

The zircons isolated from this sample are typically colourless, pale yellow, dark brown or black, euhedral and slightly elongate, and between 50 × 80 µm and 160 × 80 µm in size. Most grains have faint internal zonation, whereas the remainder are internally structureless. Many grains have a speckled appearance in transmitted light, and many are metamict. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 13 April 2003. The counter deadtime was 24 ns. Ten analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 1.63% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 3.1, 6.1, 8.1, 13.1, 15.1, and 23.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Twenty-four analyses were obtained from 24 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with a recent episode of radiogenic-Pb loss. On the basis of their $^{207}Pb/^{206}Pb$ ratios, most analyses can be assigned to one of two groups. Nineteen concordant to highly discordant analyses of 19 zircons, assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2660 ± 4 Ma (chi-squared = 1.17). Concordant to slightly discordant analyses 4.1, 17.1, 18.1, and 22.1, assigned to Group 2, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2684 ± 11 Ma (chi-squared = 0.56). Discordant analysis 14.1 indicates a significantly older $^{207}Pb/^{206}Pb$ date than these populations.

The date of 2660 ± 4 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 19 concordant to highly discordant analyses of Group 1 is interpreted as the age of igneous crystallization of the monzogranite. The analyses of Group 2 and analysis 14.1 are interpreted to be of xenocryst zircons.

Recommended reference for this publication:

NELSON, D. R., 2005, 178058: biotite monzogranite, No. 1 Bore; Geochronology dataset 585; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 13/04/2003; Data released: 30/06/2005

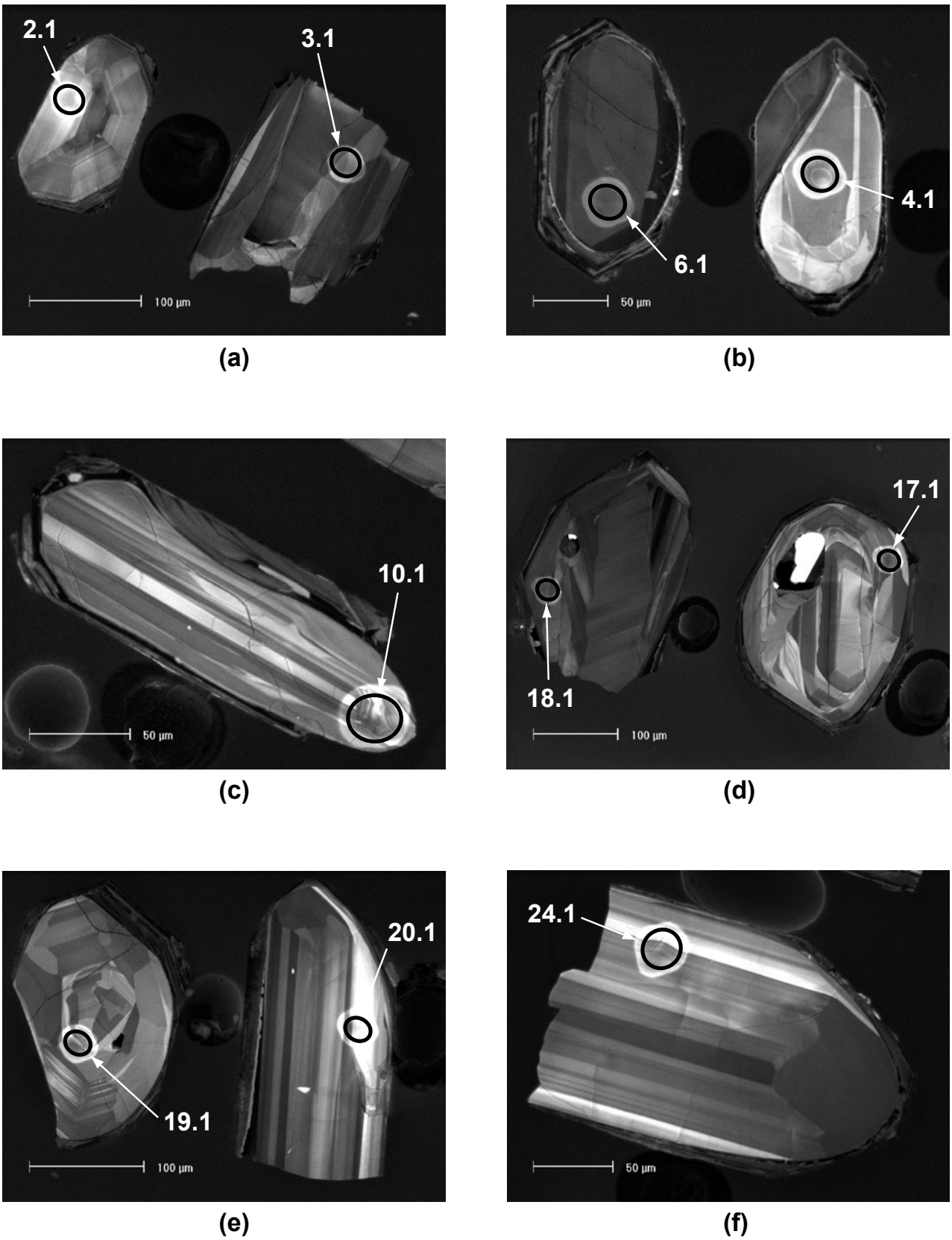

Figure 1. Cathodoluminescence images of representative zircons from sample 178058: biotite monzogranite, No. 1 Bore

Table 1. Ion microprobe analytical results for zircons from sample 178058: biotite monzogranite, No. 1 Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	105	221	74	0.673	0.18185	0.00106	0.45841	0.00247	0.4901	0.0082	12.288	0.226	96	2 670	10
2.1	93	76	58	0.130	0.17963	0.00089	0.22601	0.00156	0.5115	0.0086	12.667	0.229	101	2 649	8
3.1	104	125	64	1.823	0.17813	0.00150	0.35576	0.00327	0.4377	0.0073	10.751	0.211	89	2 636	14
4.1	96	258	80	0.055	0.18367	0.00089	0.73594	0.00269	0.5040	0.0084	12.762	0.229	98	2 686	8
5.1	84	81	53	0.086	0.18246	0.00096	0.27007	0.00176	0.5047	0.0085	12.696	0.231	98	2 675	9
6.1	315	240	182	0.913	0.18214	0.00068	0.11767	0.00124	0.4994	0.0082	12.542	0.217	98	2 672	6
7.1	227	200	140	0.063	0.18028	0.00056	0.24227	0.00094	0.5015	0.0083	12.466	0.214	99	2 655	5
8.1	209	258	133	0.945	0.18101	0.00083	0.26113	0.00166	0.4957	0.0082	12.371	0.219	97	2 662	8
9.1	196	121	114	0.059	0.18145	0.00062	0.16454	0.00097	0.5030	0.0083	12.585	0.217	99	2 666	6
10.1	142	231	102	0.275	0.18039	0.00078	0.45043	0.00181	0.5062	0.0084	12.590	0.222	99	2 656	7
11.1	143	119	86	0.235	0.18093	0.00076	0.23266	0.00135	0.4875	0.0081	12.160	0.214	96	2 661	7
12.1	184	180	112	0.559	0.18064	0.00080	0.26869	0.00160	0.4792	0.0079	11.935	0.210	95	2 659	7
13.1	245	309	169	0.901	0.18041	0.00075	0.37092	0.00161	0.5040	0.0083	12.537	0.219	99	2 657	7
14.1	92	27	85	0.074	0.34768	0.00107	0.07746	0.00098	0.7491	0.0126	35.911	0.627	97	3 697	5
15.1	89	142	44	3.088	0.17793	0.00227	0.49067	0.00523	0.3210	0.0054	7.876	0.176	68	2 634	21
16.1	193	114	113	0.055	0.18129	0.00060	0.16324	0.00089	0.5047	0.0083	12.616	0.218	99	2 665	6
17.1	139	301	110	0.117	0.18388	0.00071	0.59403	0.00191	0.5151	0.0085	13.059	0.229	100	2 688	6
18.1	193	209	123	0.678	0.18383	0.00078	0.25569	0.00153	0.5044	0.0083	12.786	0.225	98	2 688	7
19.1	71	99	49	0.267	0.17965	0.00118	0.38599	0.00259	0.5111	0.0086	12.659	0.239	100	2 650	11
20.1	36	41	23	0.420	0.17976	0.00184	0.30884	0.00384	0.4856	0.0084	12.037	0.255	96	2 651	17
21.1	142	132	88	0.060	0.18009	0.00071	0.25595	0.00128	0.4976	0.0083	12.354	0.217	98	2 654	7
22.1	173	176	111	0.144	0.18265	0.00065	0.28076	0.00120	0.5081	0.0084	12.797	0.222	99	2 677	6
23.1	114	142	75	1.462	0.18108	0.00133	0.36851	0.00291	0.4702	0.0078	11.741	0.223	93	2 663	12
24.1	84	87	54	0.287	0.17930	0.00103	0.28476	0.00200	0.5028	0.0084	12.430	0.228	99	2 646	9

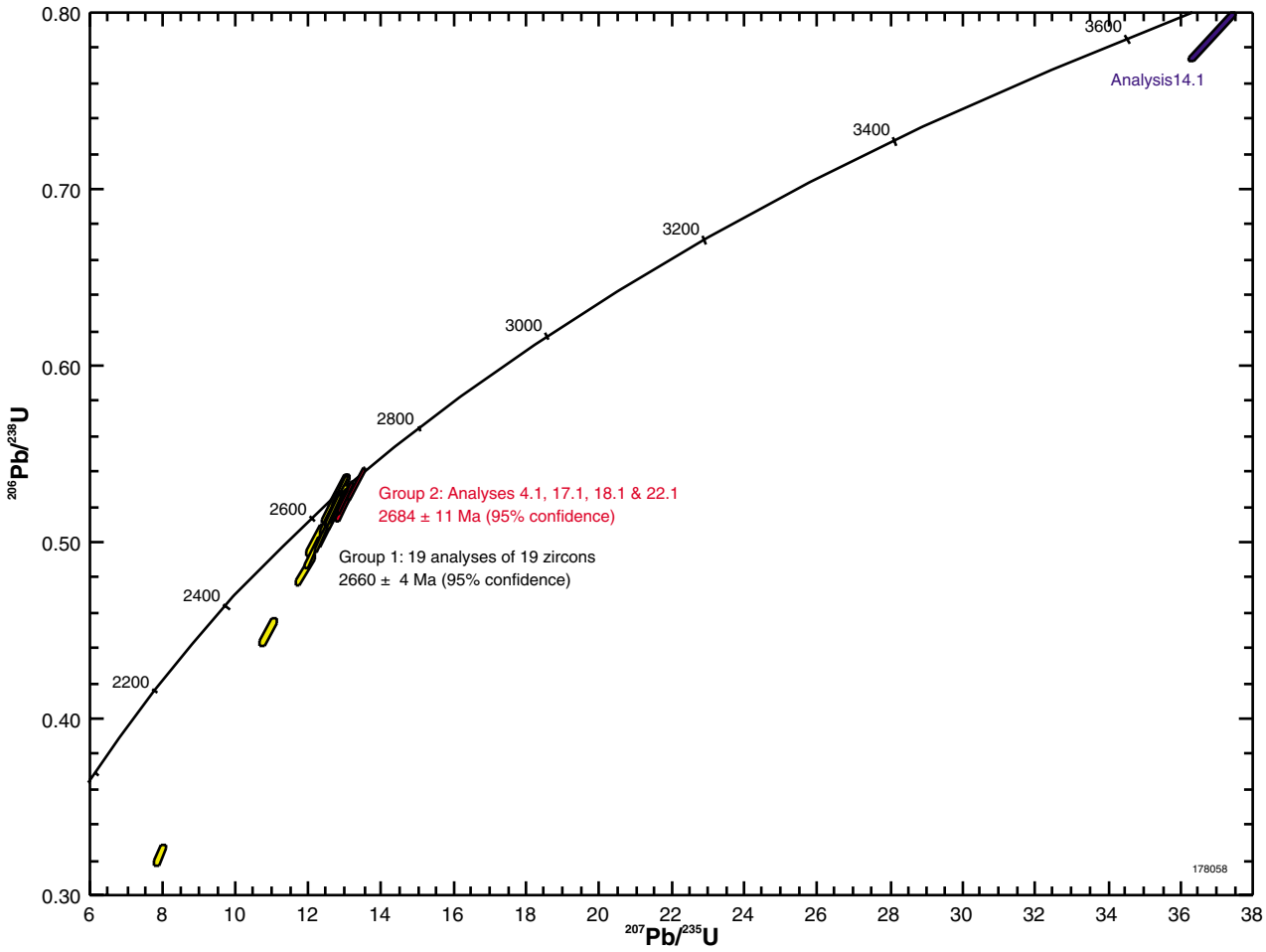

Figure 2. Concordia plot for sample 178058: biotite monzogranite, No. 1 Bore